



NASA SPACE PROJECT • EDUCATIONAL MISSION

HELION PROTOCOL

Interface brief for commanders and reviewers

What this mission is

HELION PROTOCOL is an educational recovery mission set only in our solar system. You command survey vessel HELION-7 after it comes down on Mars. There is no combat and no invented exoplanet. You walk crash-site plains, read what NASA missions actually measured, and take only the materials those worlds are known to hold.

Inner planets are landed or probed. Gas and ice giants are scooped from orbit because they have no solid surface. Earth is the industrial world — copper, silicon, oxygen, liquid water. Mars is iron oxide, perchlorate, and ice. Mercury is metal and polar frost. Venus is acid clouds over basalt at 462 °C. The giants are hydrogen, helium, ammonia, methane, and Saturn's water-ice rings.

Crafting is in-situ resource utilization: sinter a hull from hematite, split ice into propellant, build electronics from Earth, tank cryogenic hydrogen from Jupiter, and fly home on ice-giant methane. Every recipe and every survey note cites a NASA or partner dataset — MESSENGER, Magellan, Pioneer Venus, Odyssey, MRO, Phoenix, Curiosity, Perseverance, Galileo, Juno, Cassini, Voyager 2.

The solar system is the whole map. Eight planets. Nothing else.

How a commander moves

1. Crash on Mars. Survey hematite plains, polar ice, olivine, silica. Learn that perchlorate is a poison, not a supply.
2. Repair locally. Sinter a hull. Filter ice. Electrolyze the rest into a hop engine — the same ISRU idea as MOXIE, extended to fuel.
3. Use the map. Earth for copper, silicon, oxygen, water. Venus by probe. Mercury for iron-nickel and sulfides.
4. Scoop the outer system. Jupiter hydrogen, Saturn helium and ring ice, Uranus or Neptune methane. No landing.
5. Forge the return stage. Ice-giant methane plus terrestrial oxygen. Go home.

Title — the wreck

NASA SOLAR SYSTEM RECOVERY · EIGHT PLANETS ONLY

HELION PROTOCOL

Your survey ship is wrecked on Mars. Recover it with materials the planets actually hold — measured by NASA missions, not invented for a game.

Begin recovery

Project brief

AS A PLAYER

You are the commander of HELION-7. The first thing you see is not a menu of planets — it is your ship, torn open on a Martian plain at dusk. You choose to begin the recovery, or to open this project brief. As a player you are already on Mars. Nothing else in the solar system is available yet.

- Sets the crash as the only starting location.
- Project brief is the presentation deck you are reading now.
- No fictional galaxies. The subtitle states the rule: eight planets only.

Mission story — why you are here

MISSION 0 • CRASH

HELION-7 is on Mars, not in orbit.

A micrometeoroid punched the service module during the Mars capture burn. You rode the lander into a hematite plain at dusk. Hull open. Comms dark. Tanks dry.

Recovery is a science problem. Every planet in this solar system holds a different inventory — iron on Mercury, acid clouds on Venus, copper only on Earth, ice on Mars, hydrogen in the giants. NASA already measured them. You will use that list, and no other.

Land where there is ground. Probe Venus. Scoop the gas and ice giants from orbit. Craft what the wreck is missing. Then go home.



Crash site • Acidalia-class plain • Sol 0

Step onto the wreck fan

Inspect the map first

AS A PLAYER

You read the accident in plain language. A micrometeoroid during Mars capture. Hull open, comms dark, tanks dry. The briefing tells you the method before you touch a tool: land on rock worlds, probe Venus, scoop the giants, craft only from measured materials. You then step onto the wreck fan — or inspect the map first.

- Teaches access modes before gameplay: surface, probe, scoop.
- Frames recovery as a science problem, not a combat problem.

Solar map — travel the eight planets

HELION-7
PROTOCOL

Map

Survey

Cargo

Forge

Tasks

Atlas

LOC Saturn FUEL 11 SYS 6/8

HELIOCENTRIC MAP

Solar system

Distances labeled in AU. Sizes not to scale.











SUN

Mercury
0.387 AU

Venus
0.723 AU

Earth
1 AU

Mars
1.524 AU

Jupiter
5.203 AU

Saturn
9.537 AU
YOU

Uranus
19.191 AU

Neptune
30.069 AU



GAS GIANT · SCOOP

Saturn

Hydrogen-helium envelope plus the solar system's ice-particle rings. Cassini spent thirteen years measuring both.

NASA Cassini

On station

Open survey

Fuel on board 11

ACTIVE TASK

Scoop the giants

No landing. Orbit Jupiter for hydrogen, Saturn for helium and ring ice, then tank it cryogenically.

- Travel to Jupiter
- Travel to Saturn
- Craft the cryogenic tank

SHIP SYSTEMS

Hull	ONLINE
Life support	ONLINE
Propulsion	ONLINE
Heat shield	ONLINE
Comms	ONLINE
Power	ONLINE
Cryo tank	DARK
Return stage	DARK

AS A PLAYER

You look along the real order of the solar system, from Mercury to Neptune, with distances in astronomical units. Sizes are not to scale — the same disclaimer NASA uses on public diagrams. Locked worlds stay muted until a task clears them. You select a planet, read what it actually is, and spend fuel to transfer. You cannot fly until you have electrolyzed Martian ice into a hop engine.

- Only the eight planets. No exoplanets, no dwarf-planet sandbox.
- Fuel cost scales with ΔAU , so Neptune is expensive on purpose.
- Saturn's rings travel with Saturn; they are not a ninth destination.

Survey — extract what the planet has

HELION-7
PROTOCOL

Map

Survey

Cargo

Forge

Tasks

Atlas

LOC Mars FUEL 11 SYS 6/8



SURFACE OPERATIONS

Mars

Thin CO2 air. Dust, cold, and perchlorate soils. Polar ice is the local water source.

Gravity
3.71 m/s²

Temperature
-153 to 20 °C

Atmosphere
CO2 95%, N2, Ar

SITES

Survey & extract

Change planet

HELION-7 wreck fan

Olivine ridge

Polar ice scarp

Perchlorate flats

HELION-7 wreck fan

The lander sits on a hematite-stained basaltic plain typical of Meridiani-class terrain. Opportunity showed such spherules and outcrops are iron oxide, not soil dye alone.

NASA MER Opportunity; MGS TES hematite maps

Hematite ×2

Basaltic rock ×1

Logged

Exhausted

ACTIVE TASK

Scoop the giants

No landing. Orbit Jupiter for hydrogen, Saturn for helium and ring ice, then tank it cryogenically.

- Travel to Jupiter
- Travel to Saturn
- Craft the cryogenic tank

SHIP SYSTEMS

Hull	ONLINE
Life support	ONLINE
Propulsion	ONLINE
Heat shield	ONLINE
Comms	ONLINE
Power	ONLINE
Cryo tank	DARK
Return stage	DARK

You can only extract on Mars. Other worlds: Mercury, Venus, Earth, Jupiter, Saturn, Uranus, Neptune.

AS A PLAYER

You work one world at a time. Each site is a real geologic or atmospheric station: a hematite fan, a polar ice scarp, a Venus cloud-deck probe, a Jupiter envelope scoop. You log the survey first — that is the science — then extract. The yield list is the planet's measured inventory. If the site is toxic (perchlorate, sulfuric acid, 462 °C), the interface says so and will not let you pretend otherwise.

- You cannot extract from a planet you have not travelled to.
- Venus is a probe. Giants are scoops. No fake solid ground.
- Every note cites a NASA or partner instrument.

Cargo — the hold as a lab notebook

HELION-7
PROTOCOL

MapSurveyCargoForgeTasksAtlas

LOC SaturnFUEL 11SYS 6/8

HOLD

Cargo

Every unit here was taken from a solar-system body that actually contains it. Origin worlds are listed against NASA mission chemistry, not a loot table.

Basaltic rockSi-Fe-Mg-Al-Ca2

The most common volcanic crust on Venus, Earth, and Mars — tholeiitic basalt sampled by Venera, terrestrial geology, and Mars rovers.

Known on Venus · Earth · Mars

Carbon / organicsC1

Graphite, carbonates, and organic carbon. Earth is the only planet with a surface carbon cycle and abundant biomass carbon.

Known on Earth

CopperCu2

A conductive crustal metal abundant in Earth's ore deposits. No comparable copper ore has been confirmed on other planets.

Known on Earth

HeliumHe2

The second gas of the giant planets (~10% of Jupiter), Cassini and Galileo measured mixing ratios in situ.

Known on Jupiter · Saturn · Uranus · Neptune

HematiteFe2O32

Iron oxide that tints Mars red. Mapped from orbit (TES, CRISM) and identified in situ by Opportunity and Curiosity CheMin.

Known on Mars

Hydrated silicaSiO2 · nH2O1

Opaline silica and related phases found by Spirit in Gusev crater, evidence of past water-rock interaction on Mars.

Known on Mars

ACTIVE TASK

Scoop the giants

No landing. Orbit Jupiter for hydrogen, Saturn for helium and ring ice, then tank it cryogenically.

Travel to Jupiter

Travel to Saturn

Craft the cryogenic tank

SHIP SYSTEMS

HullONLINE

Life supportONLINE

PropulsionONLINE

Heat shieldONLINE

CommsONLINE

PowerONLINE

Cryo tankDARK

Return stageDARK

AS A PLAYER

You inspect mass, formula, and origin. Hematite is Mars. Copper is Earth. Hydrogen is a giant-planet gas. The hold is not a generic inventory; it is a packing list of solar-system chemistry. As a player you start to see why a hull tile and a radio cannot come from the same rock.

- Each entry names every planet known to carry that species.
- Perchlorate can sit in the hold as a hazard sample — it is never a life-support ingredient.

Forge — craft across worlds

HELION-7

PROTOCOL

Map

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Atlas

LOC Saturn FUEL 11 SYS 6/8

ISRU BENCH

Forge

Build ship systems from planetary feedstock. Recipes follow in-situ resource utilization ideas NASA already studies — sintered regolith, ice electrolysis, giant-planet scooping, methane/oxygen stages.

Sintered hull patch

ONLINE

Built

Fire hematite and basalt into a ceramic-metal skin for the torn lander.

Mars ISRU studies treat iron-oxide dust and basaltic regolith as feedstock for sintered tiles and metal extraction.

Hematite 2/2

Basaltic rock 2/1

Water-loop cartridge

ONLINE

Built

Melt polar ice and filter it through hydrated silica. Do not admit perchlorate.

Phoenix and Odyssey proved the ice. Spirit's silica is a natural filter analog. Perchlorate remains a contamination hazard.

Water ice 2/2

Hydrated silica 1/1

ISRU hop engine

ONLINE

Built

Electrolyze Martian water ice into hydrogen and oxygen bipropellant.

NASA ISRU roadmaps (MOXIE is the oxygen half) treat water ice as the local propellant source for Mars ascent and short transfers.

Water ice 2/3

Acid-ceramic heat shield

ONLINE

Built

Bind Venus sulfuric products into a high-temperature ceramic over basalt cloth.

Venus aerobot studies treat the cloud deck as a chemistry lab. The 162 °C surface forces a shield before

ACTIVE TASK

Scoop the giants

No landing. Orbit Jupiter for hydrogen, Saturn for helium and ring ice, then tank it cryogenically.

Travel to Jupiter

Travel to Saturn

Craft the cryogenic tank

SHIP SYSTEMS

Hull

ONLINE

Life support

ONLINE

Propulsion

ONLINE

Heat shield

ONLINE

Comms

ONLINE

Power

ONLINE

Cryo tank

DARK

Return stage

DARK

AS A PLAYER

You build the ship from other people's planets. A hull from Martian hematite. Propellant from ice. A heat shield that required a Venus probe. A comms array that required Earth's copper and Mercury's iron. A cryogenic tank from Jupiter hydrogen, Saturn helium, and ring ice. The return stage burns ice-giant methane with terrestrial oxygen. Missing stock is listed in the open; you go get it.

- Recipes follow NASA ISRU logic (sintered regolith, ice electrolysis, CH₄/LOX).
- You cannot substitute a material the destination does not have.

Tasks — the recovery protocol

HELION-7
PROTOCOL

Map
Survey
Cargo
Forge
Tasks
Atlas

LOC Mars FUEL 11 SYS 6/8

RECOVERY PROTOCOL

Tasks

Nine beats from wreck to return. Each one teaches a real planetary constraint — ice as fuel, Venus as a probe problem, giants as atmospheres you cannot stand on.

01 Survey the wreck

DONE

HELION-7 is down on Mars. Walk the fan, sample the oxide dust, and log the site.

You step onto rust-colored basalt and take the first scientific sample of the mission.

- Survey the wreck fan
- Collect 2 hematite

02 Close the hull

DONE

Sinter a patch from local iron oxide and basalt before the next dust storm.

You fire Martian dust into a tile and bolt it over the torn skin.

- Craft the hull patch

03 Drink the pole

DONE

Drive the scarp for ice, then the ridge for silica. Melt, filter, keep perchlorate out of the loop.

You learn that Mars has water — frozen — and that some of its salts are poison.

- Survey the polar ice scarp
- Survey the olivine ridge
- Craft the water-loop cartridge

04 Make propellant

DONE

Electrolyze the remaining ice into a hop engine. This is how you leave Mars.

You split water into hydrogen and oxygen — the same ISRU idea behind MOXIE, extended to fuel.

- Craft the ISRU hop engine

ACTIVE TASK

Scoop the giants

No landing. Orbit Jupiter for hydrogen, Saturn for helium and ring ice, then tank it cryogenically.

- Travel to Jupiter
- Travel to Saturn
- Craft the cryogenic tank

SHIP SYSTEMS

Hull	ONLINE
Life support	ONLINE
Propulsion	ONLINE
Heat shield	ONLINE
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Power	ONLINE
Cryo tank	DARK
Return stage	DARK

AS A PLAYER

You follow nine tasks, one unlocked at a time. Survey the wreck. Close the hull. Drink the pole. Make propellant. Call at Earth. Drop a Venus probe. Mine Mercury. Scoop the giants. Finish with methane. Each card tells you what you are doing as a commander and what you are supposed to notice as a student of the solar system.

- Inner planets unlock after the hop engine.
- Outer planets unlock after comms and power exist — you do not wander the Kuiper belt.

Atlas — the NASA notebook

HELION-7

PROTOCOL

Map

Survey

Cargo

Forge

Tasks

Atlas

LOC Jupiter FUEL 11 SYS 6/8

SCIENCE ATLAS

Planetary materials

A field notebook drawn from NASA Solar System Exploration fact sheets and mission instruments. If a material is not on this page, it is not in the game.

Mercury

Venus

Earth

Mars

Jupiter

Saturn

Uranus

Neptune



Jupiter

A failed star of hydrogen and helium. Galileo's probe and Juno measured the envelope that you can harvest only from orbit.

Heliocentric	Radius
5.203 AU	69,911 km
Gravity	Access
24.79 m/s²	scoop
Temperature	Atmosphere
-108 °C at 1 bar	H2 ~90%, He ~10%, NH3 clouds

NASA Galileo Probe; Juno

Confirmed feedstock

Molecular hydrogen H2

About 90 percent of Jupiter and the leading gas of Saturn, Uranus, and Neptune. Harvested only by atmospheric scooping.

Helium He

The second gas of the giant planets (~10% of Jupiter). Cassini and Galileo measured mixing ratios in situ.

Ammonia NH3

Condenses into Jupiter and Saturn's upper cloud decks. Detected by Galileo, Juno, and Cassini.

ACTIVE TASK

Scoop the giants

No landing. Orbit Jupiter for hydrogen, Saturn for helium and ring ice, then tank it cryogenically.

- Travel to Jupiter
- Travel to Saturn
- Craft the cryogenic tank

SHIP SYSTEMS

Hull	ONLINE
Life support	ONLINE
Propulsion	ONLINE
Heat shield	ONLINE
Comms	ONLINE
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Cryo tank	DARK
Return stage	DARK

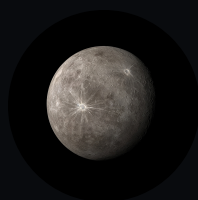
AS A PLAYER

You open the atlas when you want the argument, not the next button. Each planet page lists access mode, temperature, atmosphere, gravity, and every feedstock allowed in the game, with the mission that measured it. If a substance is not on this page, the forge will never ask for it. This is the contract with NASA data.

- Sources include MESSENGER, Magellan, Pioneer Venus, Odyssey, MRO, Phoenix, Curiosity, Perseverance, Galileo, Juno, Cassini, Voyager 2.
- Figures (AU, radii, gravity) follow NASA Solar System Exploration fact sheets.

Materials by planet

Feedstock is limited to species reported by NASA missions or standard fact sheets. Gas giants have no surface sites.



Mercury 0.387 AU · surface

A metal-rich world. MESSENGER found a bulk iron fraction far higher than Earth's, sulfur-rich surface chemistry, and water ice in permanently shadowed polar craters.

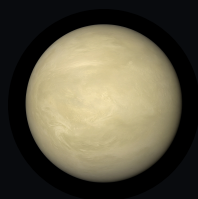
NASA MESSENGER XRS/GRS; Arecibo & MESSENGER polar ice

Iron-nickel metal · Fe-Ni

Sulfides · S / FeS

Magnesium silicates · Mg-Fe silicates

Water ice · H₂O (s)



Venus 0.723 AU · probe

A runaway greenhouse. Pioneer Venus and Magellan mapped a dense CO₂ envelope, sulfuric acid clouds, and a basaltic volcanic crust.

NASA Pioneer Venus; Magellan; Venera lander chemistry

Carbon dioxide · CO₂

Sulfuric acid aerosol · H₂SO₄

Basaltic rock · Si-Fe-Mg-Al-Ca

Nitrogen · N₂



Earth 1 AU · surface

The reference habitable world. Unique in the solar system for oceans, a biosphere, and refined crustal metals such as copper.

NASA Earth Observatory; USGS crustal abundances

Liquid water · H₂O (l)

Oxygen · O₂

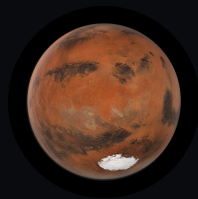
Nitrogen · N₂

Copper · Cu

Carbon / organics · C

Silicon · Si

Basaltic rock · Si-Fe-Mg-Al-Ca



Mars 1.524 AU · surface

The crash world. Iron oxides stain the dust. Rovers and orbiters have catalogued hematite, olivine, hydrated silica, perchlorates, and water ice.

NASA Odyssey, MRO, Phoenix, Curiosity CheMin, Perseverance

Hematite · Fe2O3

Olivine · (Mg,Fe)2SiO4

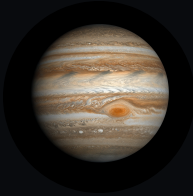
Hydrated silica · SiO2 · nH2O

Perchlorate salts · ClO4-

Water ice · H2O (s)

Carbon dioxide · CO2

Basaltic rock · Si-Fe-Mg-Al-Ca



Jupiter 5.203 AU · scoop

A failed star of hydrogen and helium. Galileo's probe and Juno measured the envelope that you can harvest only from orbit.

NASA Galileo Probe; Juno

Molecular hydrogen · H2

Helium · He

Ammonia · NH3



Saturn 9.537 AU · scoop

Hydrogen-helium envelope plus the solar system's ice-particle rings. Cassini spent thirteen years measuring both.

NASA Cassini

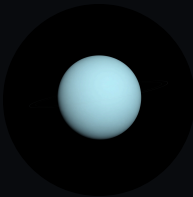
Molecular hydrogen · H2

Helium · He

Methane · CH4

Ammonia · NH3

Ring water ice · H2O (s)



Uranus 19.191 AU · scoop

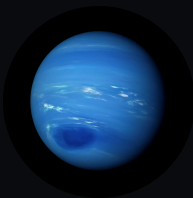
An ice giant. Methane absorbs red light and leaves the pale cyan color Voyager 2 recorded in 1986.

NASA Voyager 2

Molecular hydrogen · H2

Helium · He

Methane · CH4



Neptune 30.069 AU · scoop

The farthest planet. Voyager 2 found an active methane-stained atmosphere and the highest winds measured in the solar system.

NASA Voyager 2

Molecular hydrogen · H2

Helium · He

Methane · CH4

Forge recipes

Sintered hull patch

Mars ISRU studies treat iron-oxide dust and basaltic regolith as feedstock for sintered tiles and metal extraction.

Hematite ×2 + Basaltic rock ×1

Water-loop cartridge

Phoenix and Odyssey proved the ice. Spirit's silica is a natural filter analog. Perchlorate remains a contamination hazard.

Water ice ×2 + Hydrated silica ×1

ISRU hop engine

NASA ISRU roadmaps (MOXIE is the oxygen half) treat water ice as the local propellant source for Mars ascent and short transfers.

Water ice ×3

Acid-ceramic heat shield

Venus aerobot studies treat the cloud deck as a chemistry lab. The 462 °C surface forces a shield before any inner-planet aerocapture.

Sulfuric acid aerosol ×1 + Basaltic rock ×1 + Hydrated silica ×1

Deep-space comms array

Copper conductors and silicon electronics have no demonstrated ore-grade source on Mars. Earth remains the industrial node of the system.

Copper ×2 + Silicon ×1 + Iron-nickel metal ×1

Metal-sulfide power cell

MESSENGER's Fe-S chemistry is a natural battery analog. Carbon from Earth completes the electrode set.

Iron-nickel metal ×2 + Sulfides ×2 + Carbon / organics ×1

Cryogenic fuel tank

Giant-planet envelopes are the solar system's hydrogen well. Saturn ring ice is a convenient cryogenic heat sink.

Molecular hydrogen x3 + Helium x2 + Ring water ice x1

Odyssey return stage

A methane/oxygen stage is a NASA-relevant architecture (see Raptor-class CH₄/LOX). The methane here is planetary, not drilled.

Methane x3 + Oxygen x2 + Molecular hydrogen x1

HELION PROTOCOL · solar system only · materials after NASA PDS / Solar System Exploration fact sheets and the cited flight instruments. Distances and physical data: NASA planet compare.